Original Van der Pol Oscillation Equation

$$\frac{d^2x}{dt^2} - \mu(1 - x^2)\frac{dx}{dt} + x = 0$$

System of first order linear ODEs for the Van der Pol oscillator

$$\frac{dx}{dt} = y$$

$$\frac{dy}{dt} = \mu(1 - x^2)y - x$$

What does it model?

The Van der Pol Oscillator is an oscillating system with non-linear damping. There are two variables x and t. Where x is the position, and t represents time.

Van der Pol found stable oscillations in electrical circuits. The Van der Pol oscillator was then proposed. The second order differential equation is used in both physics and biology in various forms.

Meaning of the parameter?

There is one parameter, μ. μ is a scalar that indicates the strength of the damping and the nonlinearity.

At μ = 0, there is no damping function and the equation is just a form of the simple harmonic oscillator. At μ > 0, the equation approaches a stable limit cycle. With the exception of when x = 0. For large numbers of μ, there is a relaxed oscillation.

Cite:

Wikipedia. *Van der Pol oscillator*. https://en.wikipedia.org/wiki/Van_der_Pol_oscillator